Topic Modelling, Sentiment Polarity and Bitcoin Volatility

EDAB 6808 Assignment 4

Mamello Malape (2023232750)

Reatile Seekoei (2021109463)

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1 Executive Summary

The purpose of this assignment is to identify the dominant themes in Bitcoin-related tweets and determine whether their estimated sentiment contributes to next-day Bitcoin volatility modelling. The two questions require different uses of text: topic modelling summarises recurring discussion, whereas sentiment analysis estimates bullish, bearish or neutral outlook and tests its incremental forecasting value.

The prescribed corpus was retained throughout. Of 4,689,348 successfully parsed records, 109 were excluded for unusable dates or text, leaving 4,689,239 tweets. Topic fitting used 26,000 unique English texts, with equal monthly representation. The final NMF model produced higher coherence and keyword diversity than LDA. Trading and market discussion represented 42.13% of this balanced sample, followed by NFTs/metaverse/meme coins at 22.33% and general crypto news at 15.12%. These training proportions must be distinguished from the full-corpus monthly shares.

Question 2 used the original corpus with sentiment-specific preparation. CryptoBERT was applied to a reproducible daily sample of 40,083 tweets rather than every tweet. The 100-tweet manual review yielded 43% agreement and exposed misclassification of factual alerts and conditional market language. Daily sentiment was therefore treated as a measured, potentially noisy predictor.

The forecasting experiment compared linear regression and random forest using market-only, market-plus-volume and market-plus-volume-plus-sentiment inputs. On the same 40 test origins, market-only regression achieved the best MAE, while regression with volume achieved the best RMSE. Sentiment improved forest MAE by 5.622%, but forest RMSE by only 0.042%; both regression errors slightly worsened. The principal conclusion is that sentiment does not provide a consistent improvement across the chosen models and error measures.

NMF is retained as a descriptive tool and market-based forecasts as provisional benchmarks. Operational use would require verified timestamps, better sentiment measurement and robustness testing over additional chronological periods.

2 Introduction and Research Objectives

Bitcoin-related social media contains a mixture of market opinions, automated alerts, project promotion and broader cryptocurrency discussion. Treating all such text as a uniform measure of investor sentiment can conceal differences in both content and analytical purpose. This assignment therefore first asks what is being discussed and then tests whether a separate sentiment measure adds predictive information about daily Bitcoin movement.

The central research questions are:

1. What themes emerge from the supplied corpus, and how does their relative prevalence change under the supplied timestamps?
2. Does daily estimated sentiment improve next-day Bitcoin volatility forecasts after market information and collected tweet volume are controlled for?

The workflow follows the distinction between descriptive patterns and performance on unseen observations emphasised by Provost and Fawcett (2013). A coherent topic model does not, by itself, demonstrate forecasting value. Similarly, an apparently informative sentiment coefficient must be considered alongside validation error and held-out performance. The analysis is observational: associations are interpreted without claiming that tweets cause price movements.

3 Data Collection and Pre-Processing

3.1 Dataset and analytical scope

The assignment specifies the Kaggle Bitcoin Tweets Dataset 2025–2026 (pokeash, nd). The same corpus supports both questions; no older replacement dataset was introduced. Daily BTC–USD prices provide the separate market target for Question 2. Source records were imported in chunks using the C CSV parser with backslash escaping. Malformed records generated warnings and were skipped. Consequently, the parsed count is a documented analytical starting point rather than a verified count of every physical source record.

Table 1: Data preparation and analytical samples

Stage	Records	Interpretation
Successfully parsed source	4,689,348	Readable records under the agreed parser
Essential date/text exclusions	109	Unusable dates or blank original text
Retained modelling base	4,689,239	Shared source for the two questions
Non-empty topic-cleaned text	4,669,440	19,799 texts become empty after cleaning
Unique English topic sample	26,000	2,000 per supplied month
Assigned full-corpus NMF topics	4,641,399	47,840 left unassigned
Sentiment-eligible tweets	4,686,913	Question 2 eligibility differs slightly
CryptoBERT scored sample	40,083	Up to 200 tweets per observed day
Manually reviewed sample	100	Human labels used for validation
Usable forecasting origins	200	160 training and 40 test observations

These sample sizes answer different questions. Topic training estimates reusable themes; the fitted model is then applied to the full retained corpus. Sentiment inference uses a smaller daily sample for computational feasibility, while activity counts retain the full sentiment-eligible volume. The forecasting sample counts days with complete predictors and a next-day market outcome, not tweets.

3.2 Topic-text cleaning and duplicate treatment

Topic cleaning removes URLs, account mentions, retweet markers, punctuation, isolated numbers, stop words and very short tokens. Hashtag and cashtag markers are removed while their words remain. The shared terms Bitcoin and BTC are also removed because they provide little discrimination within this corpus. Original text is preserved so that interpretations can be checked against the tweets.

These choices prioritise recurring thematic vocabulary rather than sentence-level tone. They are therefore not reused unchanged for sentiment inference. In particular, removing negation or punctuation before sentiment analysis could change the meaning of a statement.

The source audit found 119,142 repeated original tweet texts but no repeated full records under the hash check. Duplicate cleaned texts were excluded from topic training so that repeated templates did not occupy multiple training positions. The full-corpus analysis retains the agreed treatment of repeated discussion. Missing optional user metadata does not automatically exclude an otherwise usable tweet; essential date and original-text fields determine the modelling base.

3.3 Language screening and the balanced topic sample

The fitted topic sample contains 26,000 unique English cleaned texts: 2,000 from each of thirteen supplied months. This limits the influence of the largest collection months when the vocabulary and themes are

learned. The earlier language-screening sample was estimated to be 97.035% English, although short-text language detection remains imperfect.

Equal monthly sampling means that the 42.13% trading share in the training sample is not automatically the trading share of all collected tweets. Training-sample shares describe the fitted sample; full-corpus monthly shares use assigned collected tweets as their denominators. The English fitting restriction also does not establish reliable topic assignments for every non-English tweet.

3.4 Coverage and timestamp credibility

The sentiment collection window extends from 6 March 2025 to 2 March 2026. There are 201 days with collected tweets and 161 unobserved days across 362 calendar days. Figure 1 shows that coverage varies substantially between months. First and last months are also partial windows. An unobserved day is a collection gap, not evidence that no Bitcoin discussion occurred.

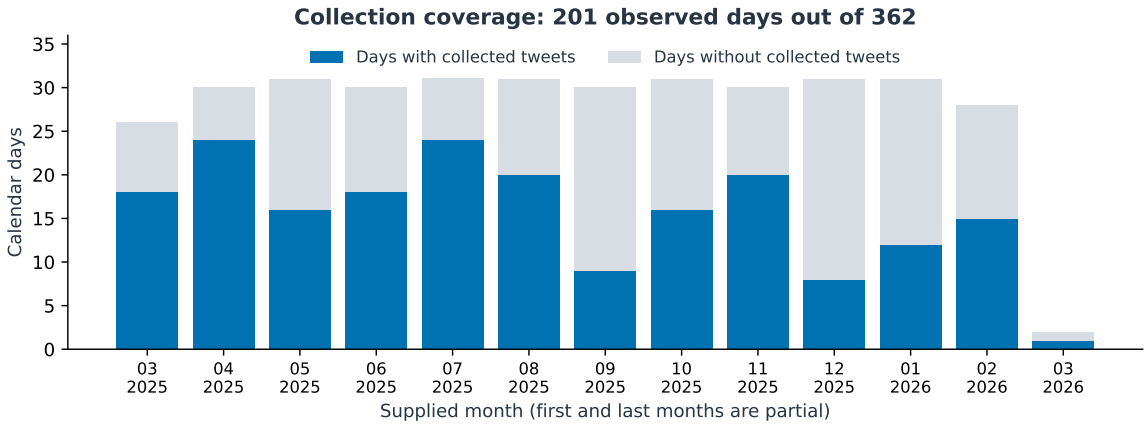


Figure 1: Collection coverage under the supplied dates. Blue bars show days with collected tweets; grey segments show unobserved days. March 2025 and March 2026 are partial collection windows.

Some original text refers explicitly to 2021–2022 while its supplied timestamp is in 2025–2026. These inconsistencies were retained and disclosed rather than repaired without evidence. They are especially important for Question 2: even correct next-day alignment in code cannot establish genuine historical tweet–market alignment when source timestamps are unreliable.

4 Text Representations and Topic-Model Selection

4.1 Count vectors, TF–IDF, LDA and NMF

Two representations and two methods were compared on the same topic sample. CountVectorizer represents term occurrences for LDA, a probabilistic topic model that represents documents as mixtures of topics (Blei et al., 2003). TF–IDF weights terms by their importance relative to the corpus and is used with NMF, which factorises the non-negative document–term matrix into document and topic weights (scikit-learn developers, nd). The comparison therefore tests both an occurrence-based and a weighted representation rather than relying on a single text encoding.

Final vectorisers use up to 5,000 unigram/bigram features, a minimum document frequency of five and a maximum document-frequency proportion of 0.95. Bigrams help preserve expressions such as “cardano price” or “free lightning”. LDA uses online learning, 15 iterations and a batch size of 256. Final NMF

uses `nndsvda` initialisation and a 500-iteration limit. Seed 42 supports reproducibility. The completed comparison does not require unused embeddings.

4.2 Evidence for selecting nine topics

The preliminary NMF assessment compared five, seven, nine, eleven and thirteen topics. Reconstruction error measures matrix fit; coherence assesses word co-occurrence; diversity measures the distinctness of the top-word sets; and stability assesses similarity across two seeds. No one diagnostic is sufficient: more components can reduce reconstruction error without producing a more useful interpretation.

Table 2: Preliminary NMF topic-count diagnostics

Topics	Reconstruction error	Coherence	Diversity	Stability
5	157.915339	-0.799463	0.960000	0.772700
7	157.102605	-1.058880	0.957143	0.710368
9	**156.469075**	**-0.956105**	**0.955556**	**0.875123**
11	155.892537	-1.131176	0.954545	0.884320
13	155.441853	-1.232112	0.953846	0.830957

Nine topics provide a compromise between detail, coherence and repeatability. Stability is substantially higher than for seven topics. Eleven topics offer slightly higher stability but worse coherence, while thirteen have worse coherence and lower stability than nine. Five topics have better coherence but less thematic detail. The preliminary and final fitting settings differ, so Table 2 supports a modelling judgement rather than proving a unique optimum. The final keywords and representative tweets provide the semantic check.

4.3 Final LDA–NMF comparison

Table 3: Final topic-model comparison on the balanced training sample

Model	Coherence	Diversity	Topic balance	Largest topic (%)
LDA	-2.275	0.767	0.953	19.892
NMF	-1.249	0.956	0.732	42.131

NMF has less-negative coherence and higher keyword diversity, supporting more distinct thematic descriptions in this sample. LDA distributes documents more evenly across topics, but equal-sized topics are not necessarily more meaningful. The lower NMF balance reflects a broad trading topic. NMF is therefore selected for interpretation and full-corpus assignment, with concentration and possible topic overlap recognised as limitations.

5 Topic Results and Change over Time

5.1 Dominant themes and interpretation

Trading/market discussion, NFTs/metaverse/meme coins and general crypto news jointly represent 79.58% of the balanced sample. The remaining topics identify airdrops, Lightning faucets, Coinhunt gaming, Cardano price updates, whale/volume alerts and mining announcements. Figure 2 shows that the fitted corpus is not restricted to direct Bitcoin price opinions.

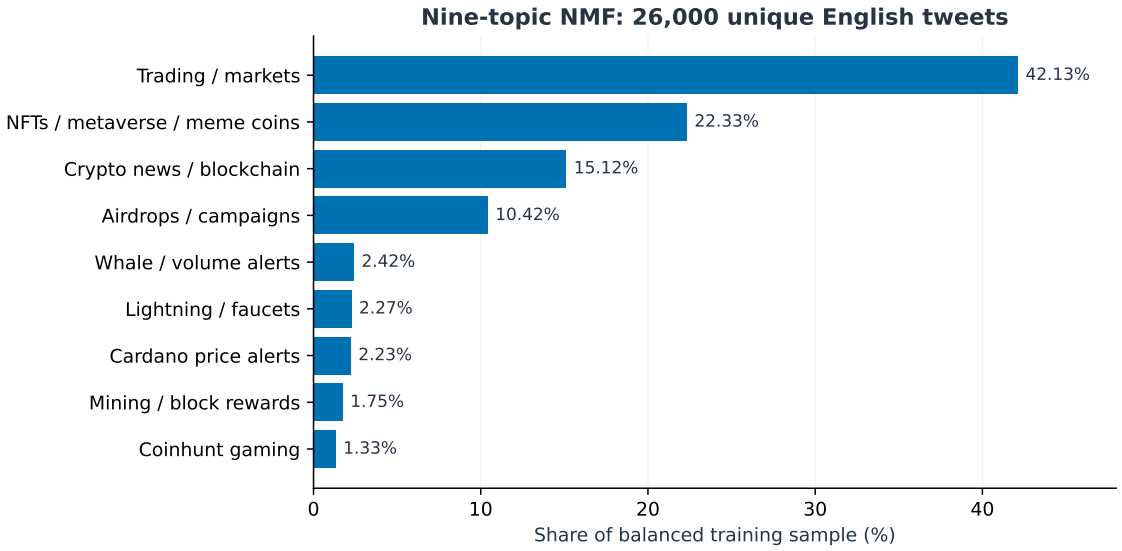


Figure 2: NMF topic shares in the balanced 26,000-tweet English training sample. These are not overall full-corpus shares.

Table 4 connects every final label to its keywords, an original representative excerpt and a concise interpretation. Excerpts come from high-scoring tweets in the same completed fit. URLs, emoji and line breaks are omitted where necessary. A high assignment score identifies an illustrative model-selected example; it does not validate every tweet assigned to that topic.

Table 4: All nine final NMF topics, keywords and illustrative tweets

Topic / sample share	Top keywords	Excerpt and interpretation
0. General crypto news and blockchain (15.12%)	crypto, cryptocurrency, blockchain	"#Bitcoin may clash again" *Interpretation:* Broad ecosystem news and cryptocurrency-project discussion.
1. Lightning network and Bitcoin faucets (2.27%)	free lightning, sats, faucet	"This zesty faucet from @_bitcoiner is making me tweet this to claim my free Lightning sats." *Interpretation:* Small Bitcoin rewards and Lightning promotions.
2. Coinhunt cryptocurrency gaming (1.33%)	coinhuntworld, vault, playing	"Carebear2614 found #bitcoin in a User vault at this location! Join me playing #coinhuntworld" *Interpretation:* Location-based gaming and reward participation.
3. Airdrops and promotional campaigns (10.42%)	project, airdrop, bsc	"@airdropinspect Good project Good project" *Interpretation:* Giveaways and repeated project-promotion wording.
4. Cardano price alerts (2.23%)	cardano price, ada usd, alert	"Current #Cardano price Alert: #ADA $0.513927 USD" *Interpretation:* Recurring other-coin price updates in the supplied corpus.
5. Trading and market discussion (42.13%)	buy, market, price, bullish, sell	"Just smashed that buy button!!!!" *Interpretation:* Trading opinions, buying/selling and price expectations.

Table 4 continued

Topic / sample share	Top keywords	Excerpt and interpretation
6. NFTs, metaverse and meme coins (22.33%)	nft, eth, metaverse, doge	"5% Rewards in Eth!! #Risingsuncoin #nftcommunity #metaverse" *Interpretation:* Wider crypto projects and promotional discussion.
7. Whale and volume alerts (2.42%)	volume, whale, alert, price	"#CFXUSDT Bear Alert! 5X Volume Price: 0.1358" *Interpretation:* Automated unusual-volume and transaction alerts.
8. Bitcoin mining and block rewards (1.75%)	block, mined, remaining	"Block 737747"; "19,048,425.00" Bitcoin "mined" *Interpretation:* Block announcements, distributed value and remaining Bitcoin supply.

5.2 Full-corpus assignment and topic changes

The selected NMF model was applied in chunks to the retained corpus, assigning 4,641,399 tweets and leaving 47,840 unassigned. Topic scores are normalised NMF weights, not calibrated probabilities. Empty or insufficiently supported text is not forced into a substantive topic. Row identifiers and supplied dates allow the assignments to be paired with the separately saved cleaned text.

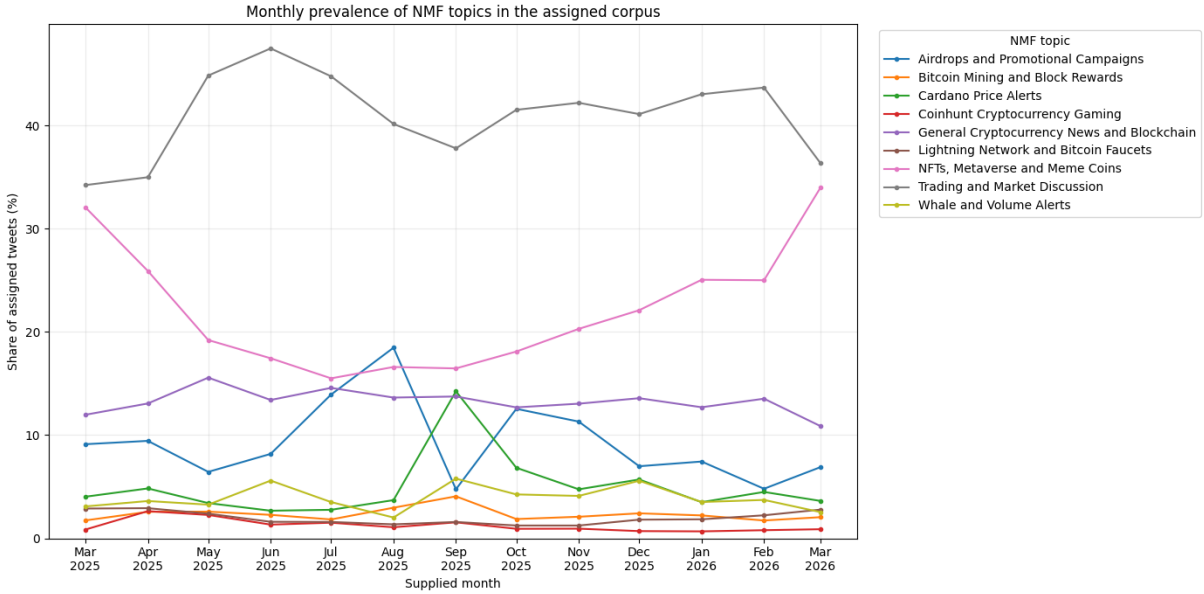


Figure 3: Monthly topic shares among assigned, collected tweets under the supplied timestamps. Partial boundary months and collection gaps limit historical comparisons.

Trading remains the largest plotted theme and has its strongest share around June 2025. The NFT, metaverse and meme-coin topic declines into mid-2025 and recovers toward early 2026. Airdrop campaigns become more prominent around July–August, while Cardano alerts show a marked September spike. These patterns suggest changing composition between broad market discussion, promotional activity and automated alerts within the collected corpus.

An exact early comparison illustrates the distinction between a share and growth: NFT share falls from 32.08% in March to 25.92% in April, while trading rises from 34.24% to 35.00%. March is a partial collection month, so this is not a like-for-like estimate of population activity. A higher topic share also need not imply more tweets in absolute terms. Later changes are read qualitatively from the saved

full-corpus chart; no unobserved numerical endpoint changes are inferred. Historical-event explanations are withheld because timestamp integrity has not been established.

6 Sentiment Polarity and Manual Validation

6.1 Sentiment preparation and decision logic

Sentiment preparation starts from original text. It preserves case, negation, punctuation, emoji and hashtags, replacing URLs with `HTTPURL` and mentions with `@USER`. This retains directional wording that topic cleaning would remove. A reproducible sample of up to 200 tweets per observed day yields 40,083 scored tweets. This daily cap supports feasible inference and daily comparison; it does not represent all 4.69 million tweets or a volume-weighted corpus census.

The fixed `ElKulako/cryptobert` model uses a 128-token limit (ElKulako, nd). Its Bearish, Neutral and Bullish classes are mapped to negative, neutral and positive for the assignment. For tweet i, polarity is

$$s_i = p_i(\text{positive}) - p_i(\text{negative}), \qquad -1 \leq s_i \leq 1. \tag{1}$$

The label is the class with the highest probability. It is not obtained by thresholding s_i at zero. For example, neutral can be the highest-probability class even when positive probability exceeds negative probability. Thus a positive polarity and a neutral label are not contradictory.

6.2 Independent manual review and classification errors

A reproducible random sample of 100 tweets was reviewed without showing model predictions. Human labels were preserved and compared with saved CryptoBERT labels. Accuracy measures agreement with those judgements, not objective truth about market direction. Human review also contains ambiguity, particularly for factual information, conditional outlook and general positive tone.

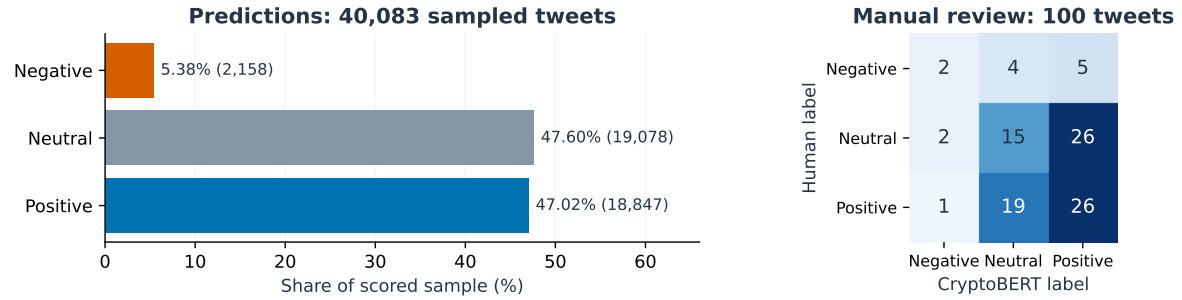


Figure 4: Predicted shares in the 40,083-tweet scored sample and the 100-tweet manual-review confusion matrix. Rows in the matrix are human labels; columns are model predictions.

CryptoBERT agrees on 43 of 100 tweets, below the 46% benchmark obtained by always predicting the most common human label. Macro-F1 is approximately 0.38. Only two of eleven human-negative tweets are predicted negative, while 26 of 43 human-neutral tweets are predicted positive. The overall scored sample has 5.38% negative, 47.60% neutral and 47.02% positive labels; these are predicted sample proportions, not verified investor sentiment.

These errors explain why domain-specific pretraining alone is insufficient evidence of good performance on this corpus. Estimated polarity is retained for the assignment, but its measurement error must accompany

Table 5: Illustrative misclassification cases from the manual review

Tweet content / row ID	Human	Model	Interpretation
"lower lows" (2209569)	Negative	Neutral	Bearish direction not recovered
Factual BTC price update (2385613)	Neutral	Negative	Price information read as bearish outlook
Whale transaction alert (1887857)	Neutral	Positive	Activity alert read as bullish outlook
Conditional long opportunity (4439250)	Positive	Neutral	Conditional bullish wording not recovered

any interpretation of sentiment–volatility relationships. No claim is made that inference accuracy improves simply because the model is called CryptoBERT.

6.3 Daily aggregation and full collected volume

For each observed UTC day, sentiment features are mean tweet polarity and the shares labelled negative and positive. Neutral share remains available for description but is omitted as an additional predictor because the three shares sum to one. Activity is the logarithm of one plus the full sentiment-eligible tweet count. The inference-sample count is not substituted for that full volume.

Days without sentiment remain missing and are excluded when complete predictors are required. They are not assigned neutral sentiment. This preserves the distinction between a measured neutral day and an unobserved collection day. The daily variables summarise sampled outlook and collected activity; they do not measure all Bitcoin investors or all messages published on the platform.

7 Volatility Target and Time-Aware Design

7.1 Bitcoin prices and the movement-size target

Daily BTC–USD close prices were obtained from Yahoo Finance using yfinance (Yahoo Finance, nd). Returns are expressed in percentage units:

$$r_t = 100 \log\left(\frac{P_t}{P_{t-1}}\right), \qquad v_t = |r_t|, \qquad y_{t+1} = v_{t+1}. \tag{2}$$

Absolute log return is a transparent proxy for the magnitude of daily movement. It is useful here because the question concerns volatility rather than a rise or fall in price. However, it is not a conditional variance estimate, a GARCH forecast or an intraday realised-volatility measure. The target choice therefore defines the practical scope of the conclusions.

7.2 Calendar alignment and the chronological split

Each forecasting origin is UTC day t, after its close and tweet information are available. Inputs from t predict the next calendar day's magnitude, y_{t+1}. Relative to the target, sentiment and market inputs are lagged by one day. The target shift is performed on the complete price calendar before filtering for sentiment availability. This prevents an unobserved tweet day from being confused with the next observed tweet day.

Of 362 calendar rows, 200 origins have complete predictors and targets. The 161 collection gaps cannot supply sentiment predictors, and the final observed sentiment day lacks a next-day price in the saved

price window. All models use the same retained origins so that missingness does not create different test samples for different input groups.

Table 6: Chronological evaluation design

Partition	Origins	Origin dates
Training (earliest 80%)	160	6 March–19 November 2025
Test (latest 20%)	40	20 November 2025–28 February 2026

Test target dates run from 21 November 2025 to 1 March 2026. Exploratory correlations and model fitting use training observations. Test values are reserved for evaluation; there is no random row split or reported test-driven parameter tuning. This implements the distinction between fitting performance and generalisation emphasised by Provost and Fawcett (2013).

7.3 Training-period exploratory relationships

Training mean daily polarity is approximately 0.408, with a standard deviation of 0.045. The training target has a mean of approximately 1.566 percentage points. Mean polarity's Pearson correlation with next-day movement is -0.019, versus -0.219 for log tweet volume and 0.162 for current absolute return. The sentiment correlations are therefore weak in this sample.

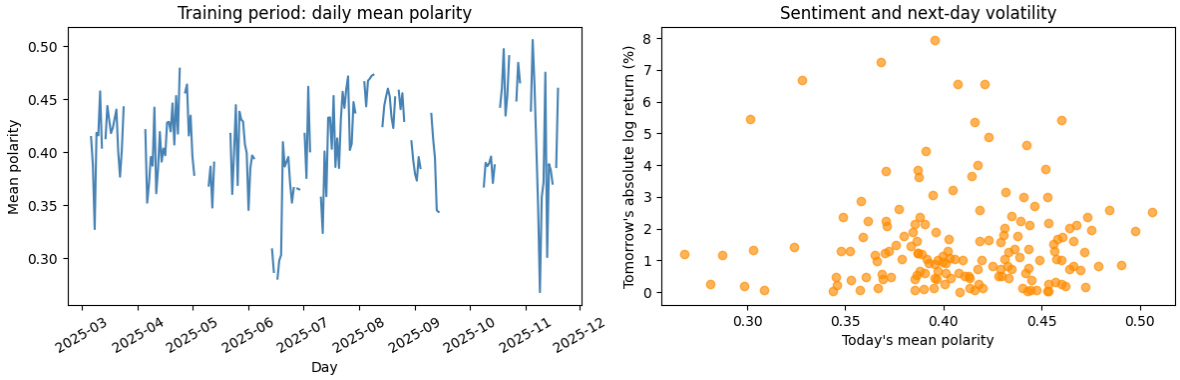


Figure 5: Training-period daily mean polarity and its association with next-day absolute Bitcoin returns. These plots are descriptive and do not identify causal effects.

Weak linear correlation does not rule out nonlinear information, which motivates including random forest as well as regression. Conversely, an apparent scatter pattern is not sufficient evidence for predictive usefulness. The notebook's exploratory stages also document sentiment distribution, supplied time coverage, collected volume, signed returns and movement magnitude. The report focuses on the evidence relevant to model comparison rather than reproducing every code cell.

8 Model Evaluation and Interpretation

8.1 Model families, input groups and benchmarks

Both linear regression and random forest were fitted to three matched input groups. This design separates the effect of collected activity from the effect of sentiment. Comparing a market-only regression with a sentiment-enhanced forest alone would confound a predictor change with a model-family change.

Table 7: Matched predictor groups used by both model families

Group	Predictors	Variables
Market	2	Current signed and absolute log returns
Market + volume	3	Market inputs and $\log(1 + \text{full tweet count})$
Market + volume + sentiment	6	Add mean polarity, negative share and positive share

The forest uses 200 trees, maximum depth four, minimum leaf size five and seed 42. Parameters remain fixed for the comparison. Six fitted models are evaluated alongside training-mean, training-median and persistence forecasts. Persistence uses today's absolute return as tomorrow's prediction. Simple benchmarks are necessary because a more complex model can still fail to improve on a constant forecast (Provost and Fawcett, 2013).

8.2 Held-out errors and overall model ranking

MAE measures the typical absolute prediction error, while RMSE gives greater weight to large misses. Both are reported in percentage points because the target uses percentage log returns. They are not classification accuracy percentages. R^2 provides a further comparison with a constant test-mean prediction; negative values indicate performance below that reference in this test sample.

Table 8: Forecast performance on the same 40 test origins

Model / input group	MAE	RMSE	R^2
Linear regression: market	**1.1340**	1.6331	0.3492
Linear regression: market + volume	1.1664	**1.5938**	0.3802
Linear regression: market + volume + sentiment	1.1715	1.5944	0.3797
Random forest: market	1.2924	2.0717	-0.0472
Random forest: market + volume	1.2262	1.8085	0.2020
Random forest: market + volume + sentiment	1.1572	1.8077	0.2027
Training mean benchmark	1.1832	2.0278	-0.0033
Training median benchmark	1.1660	2.0923	-0.0682
Persistence benchmark	1.6730	2.4669	-0.4849

Market-only linear regression has the lowest MAE at 1.1340. Regression with volume has the lowest RMSE at 1.5938. The sentiment-enhanced forest is competitive by MAE, but is not the best overall model on that measure and has substantially higher RMSE than the regression models. Persistence performs worst on both error measures. No raw forecasts are negative, so flooring predictions at zero does not change the reported results.

The ranking depends on the chosen loss measure. Adding volume to regression reduces the larger errors sufficiently to improve RMSE but slightly increases MAE. This is one reason to avoid selecting a model from a single isolated score or describing a lower error as a percentage of Bitcoin volatility explained.

8.3 Does sentiment add incremental forecasting value?

Sentiment's contribution is assessed against the market-plus-volume control within each family. Positive improvement denotes lower error after sentiment is added. The saved full-precision calculations are used; displayed error tables are rounded.

The forest's 5.622% MAE reduction indicates better average absolute errors for this split, but its almost unchanged RMSE indicates little improvement in the larger misses. Regression does not benefit on either

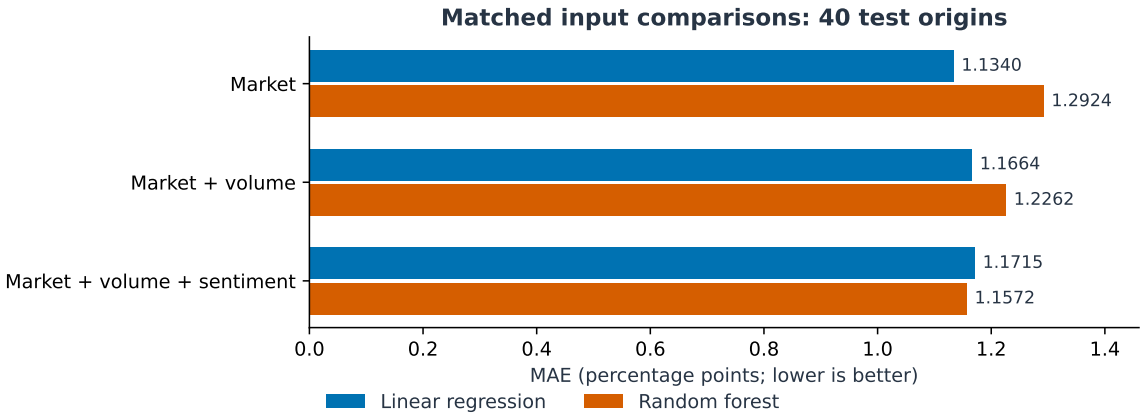


Figure 6: MAE comparison across identical predictor groups and test origins. Lower values indicate smaller errors in percentage points.

Table 9: Incremental sentiment value after controlling for market inputs and volume

Family	MAE improvement (%)	RMSE improvement (%)
Linear regression	-0.435	-0.041
Random forest	5.622	0.042

measure. Figure 7 shows almost overlapping regression forecasts and some changes in ordinary-day forest forecasts; both model families still substantially miss the largest observed movement.

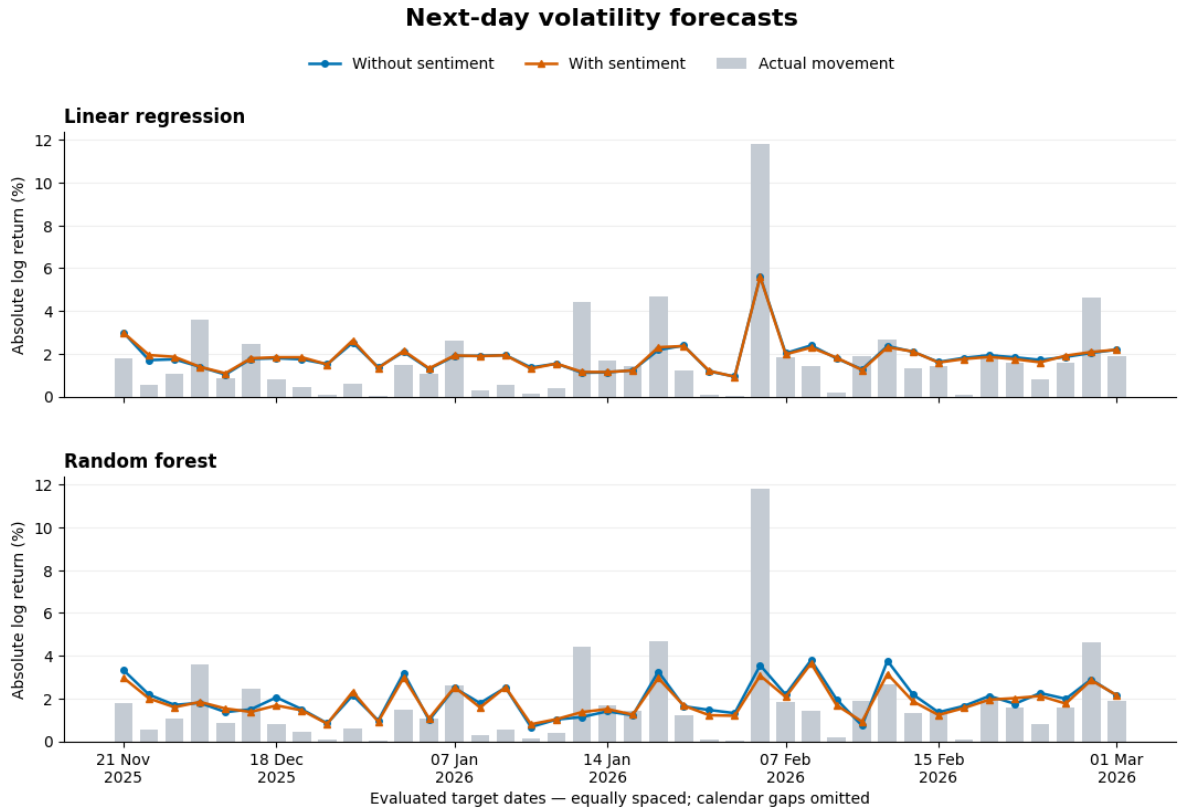


Figure 7: Actual next-day movement and matched forecasts with and without sentiment. Both comparisons include market inputs and tweet volume. The 40 test target dates are equally spaced; calendar gaps are omitted from the horizontal spacing.

The evidence therefore supports a modest, model-specific MAE gain rather than a consistent sentiment advantage. It is not a demonstration of statistical significance, profitability or robust performance across market regimes. The poor manual-validation result offers a plausible measurement concern, but this experiment does not identify the cause of the mixed forecasting results.

8.4 Feature interpretation: association rather than causation

The enhanced regression coefficients are multiplied by each predictor's training standard deviation. This expresses a conditional fitted change in predicted movement in comparable percentage-point units. Forest importance reports each predictor's share of training split importance.

Table 10: Interpretation of the two sentiment-enhanced models

Predictor	Regression change per 1 SD (pp)	Forest importance (%)
Log full tweet volume	-0.3445	36.9633
Current signed return	-0.2374	28.9317
Positive share	0.0223	14.0414
Current absolute return	0.1756	9.3698
Negative share	0.0905	5.7080
Mean polarity	0.0607	4.9857

Both positive and negative discussion can accompany greater movement magnitude: their coefficients do not imply that positive sentiment predicts rising returns or negative sentiment predicts falling returns. The target is absolute movement. Mean polarity's conditional coefficient is positive despite its near-zero univariate correlation, illustrating why fitted coefficients depend on the other inputs and should not be read as standalone causal effects.

Volume and signed return have the largest forest training importances. These values describe the fitted model's use of predictors; they are not percentages of test volatility explained. Correlated sentiment variables and noisy labels further limit an independent interpretation of their weights. Held-out changes in error provide the more direct test of added predictive value.

9 Conclusion

The assignment applied a complete text-analysis workflow to the prescribed Bitcoin-related corpus. Topic cleaning, monthly-balanced English sampling and two representation–method combinations supported the selection of nine-topic NMF. Its themes reveal a mixture of trading discussion, wider crypto projects, promotion and automated alerts, with changing composition under the supplied timestamps.

Sentiment was generated separately from original text, validated against 100 human judgements and aggregated to daily features. A calendar-aware design then compared six forecasting models and three benchmarks on identical future observations. Market-only regression achieved the lowest MAE; regression with volume achieved the lowest RMSE. Sentiment improved forest MAE but did not produce a consistent gain across models and loss measures.

The study therefore supports the usefulness of NMF for descriptive discussion analysis and retains sentiment as an experimental forecast input. The results remain conditional on source-date credibility, collection coverage, measurement quality and one small chronological test set. They support association within this analysis rather than causal or operational claims.

10 Limitations

Several limitations qualify the findings.

- **Timestamp integrity.** Textual references to earlier years conflict with supplied 2025–2026 metadata. This restricts historical event interpretation and the real-world meaning of tweet–market alignment. Retaining source dates does not validate them.
- **Incomplete coverage.** Collection gaps occur on 161 of 362 calendar days. Observed days may differ systematically from missing days, while partial boundary months complicate comparisons of monthly shares.
- **Sampling and corpus composition.** Topic fitting uses equal monthly English samples, while sentiment uses a daily cap. Neither is a population census. Repeated promotion, other-coin content and imperfect language detection can influence the results.
- **Sentiment measurement.** Agreement is only 43% in a 100-tweet, single-reviewer validation. Ambiguous language, factual alerts, multilingual text and general tone versus directional outlook complicate labelling.
- **Target and evaluation scope.** Daily absolute return captures one aspect of volatility. Only 40 origins are tested in one chronological split; no uncertainty interval or expanding-window robustness result is reported.
- **Interpretability.** Topic scores are not calibrated probabilities, and model coefficients or training importances do not establish causal effects. No independently validated trading or risk-management system is demonstrated.

11 Recommendations

NMF should be retained as the primary descriptive model because its final coherence, diversity and interpretable vocabulary support a useful summary of the supplied corpus. Market-based forecasting models should remain the provisional benchmarks: sentiment should be justified by measured incremental performance rather than assumed to be useful because it is domain-specific.

For further analytical development, the priorities are:

1. Verify source timestamps before interpreting historical market relationships or attributing topic changes to particular events.
2. Expand independent manual review, document label disagreements and improve sentiment measurement before relying on the estimated polarity in decisions.
3. Use expanding-window or repeated chronological evaluation to assess whether the modest forest MAE gain persists beyond this test period.
4. Check sensitivity to daily sampling, promotional content, language and an alternative volatility target such as a rolling standard deviation.
5. Assess stability and forecast errors around large movements, since the current sentiment enhancement does not resolve the largest miss.

These are proposed extensions, not completed results. The completed notebook, saved models, tweet-level assignments and validation outputs provide the analytical record for the present report.

12 Reference List

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